DESIGN OF 1X2 COMPACT UWB MIMO ANTENNA UISNG STAIR CASE & COMBLINE

Complete Specification

Abstract

Ultra-Wideband (UWB) communication systems require antennas that can operate across an extremely wide frequency range while maintaining good impedance matching, high efficiency, and stable radiation performance. In parallel, modern wireless technologies increasingly rely on Multiple-Input Multiple-Output (MIMO) systems to achieve higher data rates, improved channel capacity, and better link reliability. However, combining UWB and MIMO in a compact design introduces a critical challenge: when antenna elements are placed close to each other, strong mutual coupling occurs, which negatively affects isolation and overall system performance.

In this work, a compact UWB–MIMO antenna is designed and analyzed to address these challenges effectively. The proposed antenna incorporates two key structural features: a staircase-shaped radiating patch and a combline decoupling structure on the ground plane. The staircase geometry plays an important role in enhancing impedance matching and extending the operational bandwidth by generating multiple resonances. Meanwhile, the comb-line structure functions similarly to a mini electromagnetic bandgap (EBG), reducing surface current distribution and minimizing mutual coupling between antenna elements.

The antenna is modeled and simulated using ANSYS HFSS, and its performance is evaluated through various parameters such as S-parameters, bandwidth, isolation, radiation patterns, gain, efficiency, and Envelope Correlation Coefficient (ECC). The results demonstrate that the proposed design achieves wide UWB bandwidth, high isolation suitable for MIMO applications, and stable radiation characteristics across the operating range. These features make the antenna a strong candidate for next-generation UWB wireless communication systems.

Keywords: Ultra-Wideband (UWB), MIMO Antenna, Mutual Coupling, Isolation Enhancement, Staircase Patch, Comb-Line Structure, Electromagnetic Bandgap (EBG), HFSS, Envelope Correlation Coefficient (ECC), Wideband Antenna Design.

1. Introduction:

Wireless communication has become an essential part of modern life, supporting applications ranging from smartphones and IoT devices to healthcare systems and advanced sensing technologies. As the demand for faster data transmission, higher reliability, and efficient spectrum utilization continues to grow, antenna design has become a critical area of research. In particular, next-generation communication systems require antennas that are not only compact but also capable of delivering high performance over a wide frequency range.

Ultra-Wideband (UWB) technology has emerged as a promising solution to meet these requirements. By operating over a very large bandwidth, UWB systems enable high data rates, low power consumption, and strong resistance to multipath interference. These advantages make UWB suitable for applications such as short-range high-speed communication, indoor positioning, and radar sensing. However, designing antennas that can efficiently operate across the entire UWB range while maintaining stable radiation characteristics and good impedance matching remains a significant challenge.

At the same time, Multiple-Input Multiple-Output (MIMO) technology has gained widespread attention due to its ability to enhance channel capacity, improve signal reliability, and increase spectral efficiency without requiring additional bandwidth. By using multiple antenna elements at both the transmitter and receiver, MIMO systems can transmit parallel data streams and exploit multipath propagation effectively. Despite these benefits, integrating MIMO with UWB in a compact structure introduces a major issue—mutual coupling between closely spaced antenna elements. This coupling leads to reduced isolation, higher correlation, and degraded overall system performance.

To address these challenges, this work focuses on the design of a compact UWB–MIMO antenna with improved isolation and wideband performance. The proposed design incorporates a staircase-shaped radiating element to enhance bandwidth and impedance matching, along with a comb-line structured ground plane to suppress surface currents and reduce mutual coupling. By combining these techniques, the antenna achieves stable radiation characteristics, low Envelope Correlation Coefficient (ECC), and high isolation across the UWB range. The design is validated through simulation, demonstrating its suitability for modern high-speed wireless communication systems.

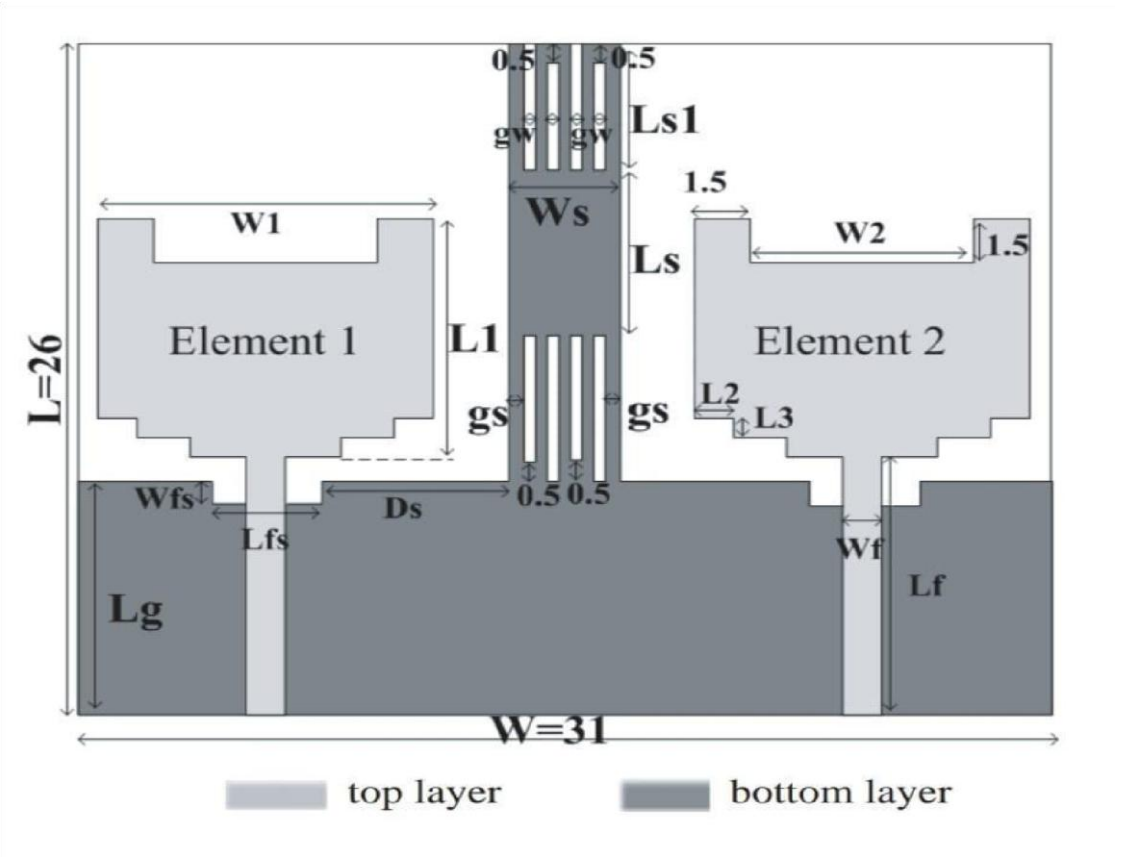


Fig. 1. Geometry of the designed antenna

L	Lfs	Lf	L_1	L_2	L_3	Lg	Ls	Ds	gs
26	4	9	8	1.5	0.5	8	7	6	0.45
W	Wfs	Wf	W_1	W_2	W_3	W_4	Ws	Ls_1	gw
31	0.8	1.4	11	8	5	8	3	5	0.3

Dimensions of the designed antenna (mm)

Need for UWB–MIMO Antenna Integration

Combining UWB and MIMO technologies provides both wide bandwidth and improved per formance, making it an ideal solution for next-generation communication systems. However, integrating multiple antennas into a compact UWB system is challenging due to mutual cou pling, which occurs when the electromagnetic field of one antenna affects the other. Negative effects of mutual coupling include: • Reduced isolation between antenna elements

- Shift in resonant frequencies
- High correlation between antennas

- Poor radiation efficiency
- Lower gain and pattern distortion
- Degraded MIMO channel performance

Therefore, special structural modifications must be used in the antenna design to minimize mu tual coupling while maintaining wideband performance.

3. UWB–MIMO Antennas in Literature:

Recent studies have focused on designs that combine radiator shaping with intelligent ground plane engineering. Researchers have explored various geometries such as fractal antennas, ta pered structures, semielliptical patches, and slot-loaded radiators to enhance bandwidth. Similarly, ground structures like combshaped slots, cross-shaped grooves, T-shaped defects, and meandered slots have been implemented to suppress surface currents. Overall, UWB–MIMO literature shows that:

- Bandwidth enhancement depends strongly on radiator geometry.
- Isolation improvement is mainly influenced by ground plane modifications.
- ECC values below 0.02 are preferred for good diversity performance.
- Compact integration is essential for modern wireless devices.
- Simple structures with fewer fabrication steps are highly preferred.

These findings highlight that combining an optimized radiator with a well-designed ground plane can significantly improve overall UWB–MIMO performance.

2. Motivation for the Proposed Design

Based on the analysis of prior research, it is clear that existing UWB–MIMO antennas face several challenges, including complexity, large size, and frequency-dependent isolation. The proposed antenna design in this project aims to overcome these challenges using two simple yet effective techniques:

- A staircase-shaped radiating patch, which introduces multiple resonances and im proves impedance bandwidth across the entire UWB range.
- A comb-line structured ground plane, which suppresses surface currents and achieves wideband mutual coupling reduction without increasing antenna size.

Both structures are easy to fabricate on FR4 substrates and do not require additional compo nents, metamaterials, or complex geometries. This makes the proposed antenna suitable for compact, low-cost UWB–MIMO systems in modern communication devices.

3. Methodology Overview

The overall methodology is based on progressively converting theoretical concepts into a fully functional HFSS model. The antenna consists of two identical printed monopole elements placed on a compact 26 × 31 mm² substrate, so the methodology emphasises both wideband impedance matching and mutual coupling reduction.

- The first part of the methodology involved understanding the antenna geometry, including U-shaped radiators, staircase corner modifications, microstrip feed lines, and ground-slot structures. These were chosen because they improve bandwidth and impedance matching over the UWB range without increasing antenna size.
- Next, detailed theoretical predictions were studied, especially lower resonant frequency estimation for planar monopoles, effects of feed gap, effective current path length, and the role of comb-line stubs. This ensured that the antenna dimensions chosen were not arbitrary but aligned with electromagnetic behaviour.
- Once the preliminary design concept was finalised, HFSS was used to transform the conceptual model into a 3D electromagnetic structure, assigning materials, ex act geometries, and applying suitable boundaries. The modelling process ensured that even small features such as the ground stubs (Ws, Ls) and ground slots (Wfs, Lfs) were implemented precisely.

- Finally, repeated simulations were carried out, comparing S-parameters, radiation characteristics, isolation behaviour, gain and efficiency until the antenna showed stable UWB bandwidth and isolation better **than –25 dB.** This methodology ensures that both UWB performance and MIMO isolation are achieved in a compact size.

Modelling Process in HFSS

The antenna design was built step-by-step inside HFSS, ensuring every structural detail from the research paper was captured accurately. HFSS's 3D modelling tools allowed precise implementation of all dimensions from Table 1 of the paper.

- **The substrate was modelled first using Rogers RO4003**, with exact length (26 mm), width (31 mm), height (0.7874 mm) and dielectric properties. This forms the fundamental platform on which all other layers are placed.
- **The two identical U-shaped radiating elements were then created on the top surface**, each consisting of a feed line (Wf × Lf), a U-shaped radiator, and the staircase structure (L2, L3). These details significantly influence lower resonant frequency and wideband behaviour.
- **The ground plane was drawn on the bottom side of the substrate**, and rectangular highfrequency matching slots (Wfs × Lfs) were cut directly beneath the feed lines.

  These slots help improve the high-frequency impedance matching.
- **The comb-line decoupling structure was modelled on the ground plane** by adding multiple stubs of dimensions (Ws × Ls) connected with a thin metal strip. This structure behaves as an electromagnetic bandgap (EBG) element and plays a key role in reducing mutual coupling. • **Each antenna element was assigned a microstrip line feed**, and the two ports were positioned such that the inter-element spacing matched the design specification.
- This modelling process created a complete and accurate 3D representation of the UWB MIMO antenna.

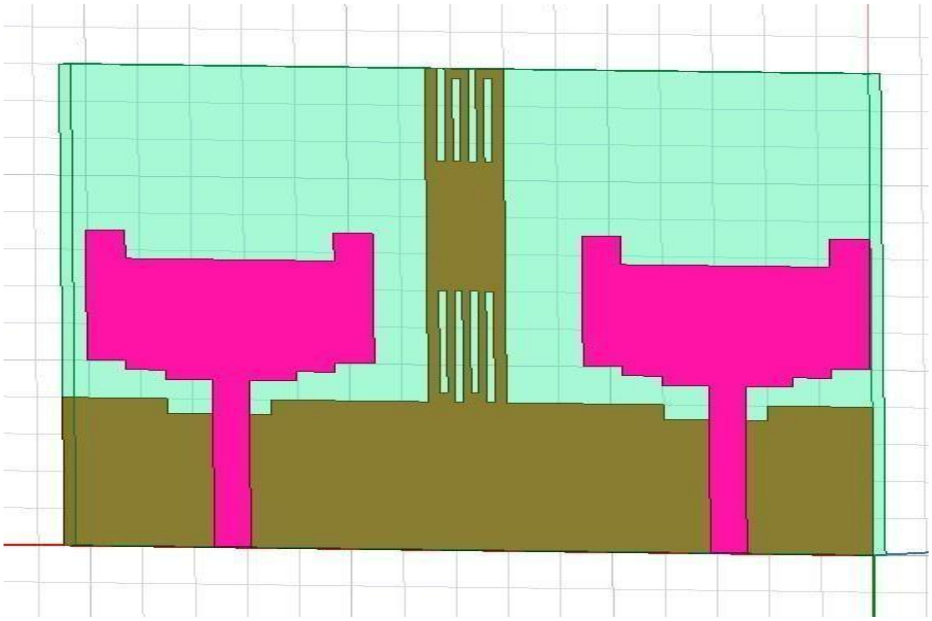


Fig 2: UWB MIMO Antenna Design

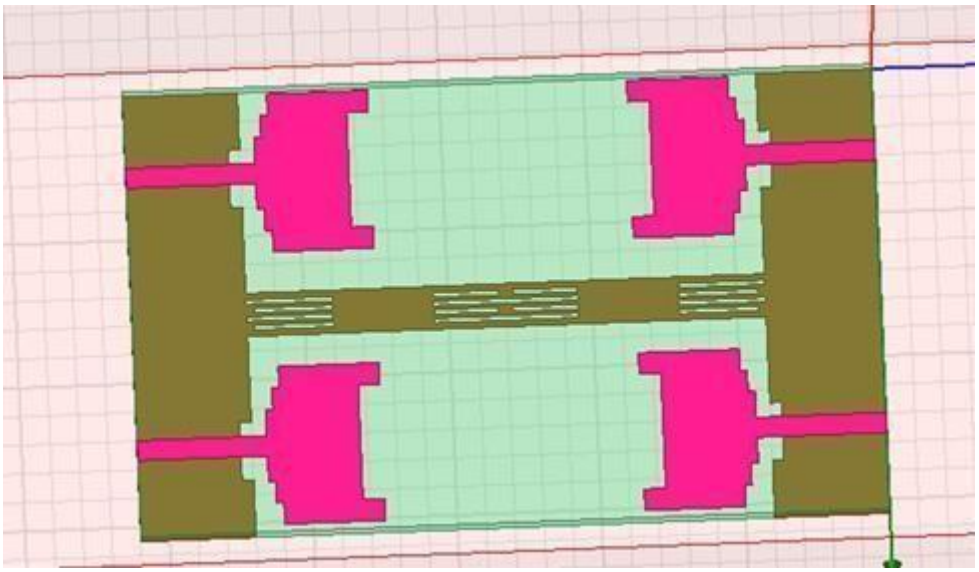


Fig 2.1: 1x2 UWB MIMO Antenna Design

4.Reflection Coefficient (S11) Analysis

The reflection coefficient (S11) is a primary indicator of how effectively an antenna receives power from its feedline. For UWB antennas, the S11 parameter must remain below –10 dB across a wide frequency range, indicating that at least 90% of the input signal is radiated rather than reflected. Key observations from S11 simulation:

- The proposed staircase radiating patch exhibits multiple resonances that merge smoothly, producing a continuous bandwidth spread across the 3.1–10.6 GHz UWB range. This verifies that the staircase geometry successfully introduces multiple resonant modes without requiring a larger physical radiator.
- The S11 curve does not show abrupt peaks or mismatches, which means the impedance transitions between different resonances are smooth. This is attributed to the overlapping effects created by the staircase edges, allowing efficient radiation across the entire spectrum.
- The partial ground plane enhances broadband matching by reducing capacitive loading between the feedline and the ground conductor. This plays a major role in maintaining stable reflection performance over a wide frequency range.
- Even after inserting the comb-line ground structure, the S11 values remain below –10 dB throughout the UWB range, demonstrating that the isolation improvement structure does not negatively impact matching performance.
- The overall S11 performance confirms that the antenna is suitable for all UWB applications including communications, radar imaging, portable devices, and indoor localization systems.

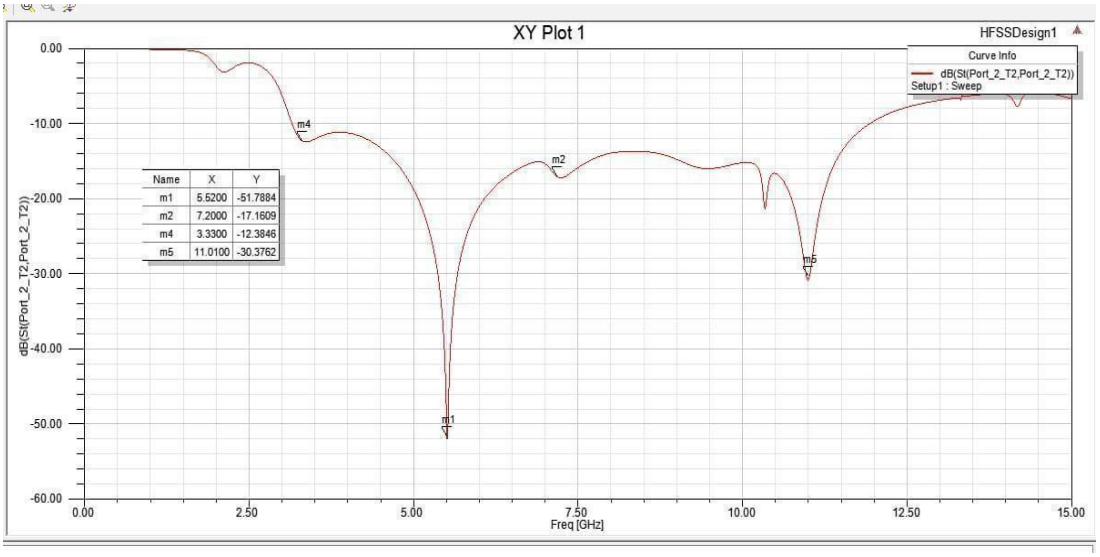


Fig 3 : S11 Parameter for first patch

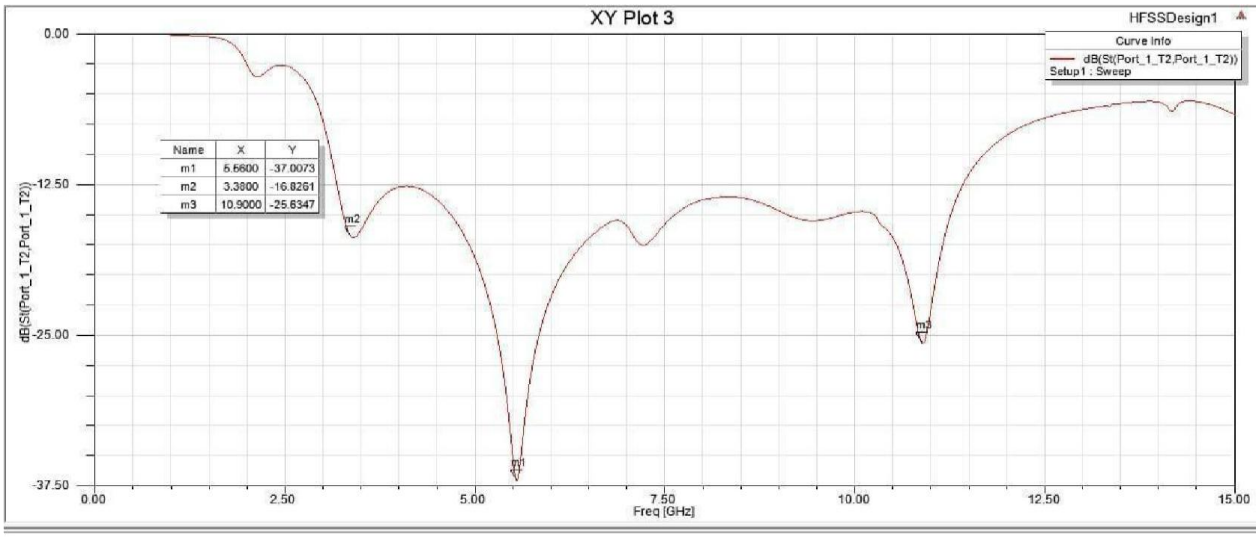


Fig 4 : S22 Parameter for second patch

Isolation (S21) and Mutual Coupling Analysis

In MIMO antennas, isolation between radiating elements is one of the most critical parameters. Since both elements operate independently, mutual coupling must be minimized to ensure good system performance. The S21 parameter indicates how much power from one antenna element leaks into the other.

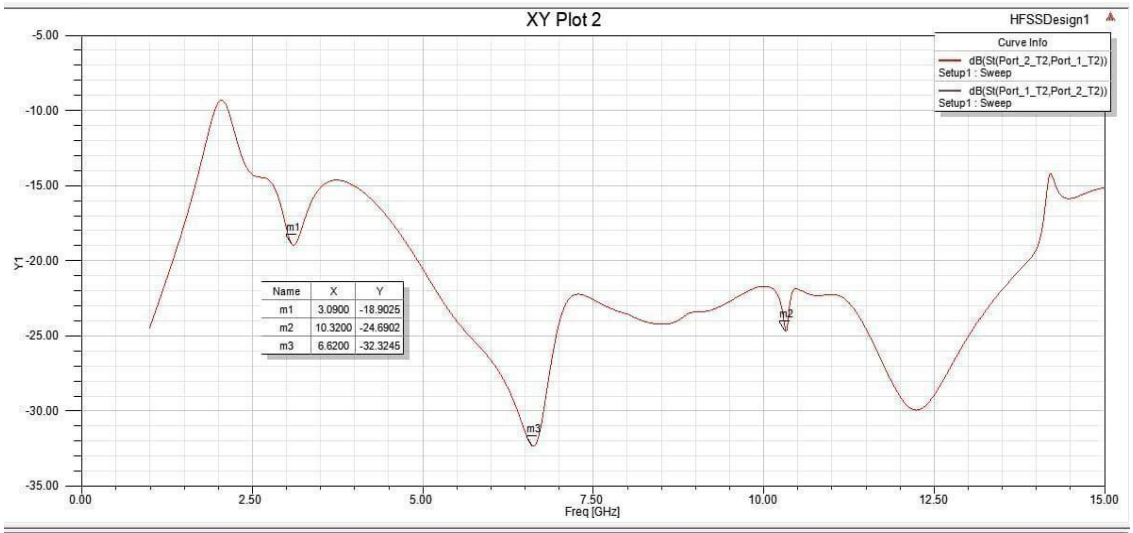


Fig 5: Isolation (S21) and Mutual Coupling Analysis

Radiation Pattern and Far-Field Characteristics

Radiation patterns describe how an antenna distributes radiated power in space. For UWB– MIMO antennas, stable and omnidirectional patterns are preferable, as they support consistent communication regardless of the device orientation.

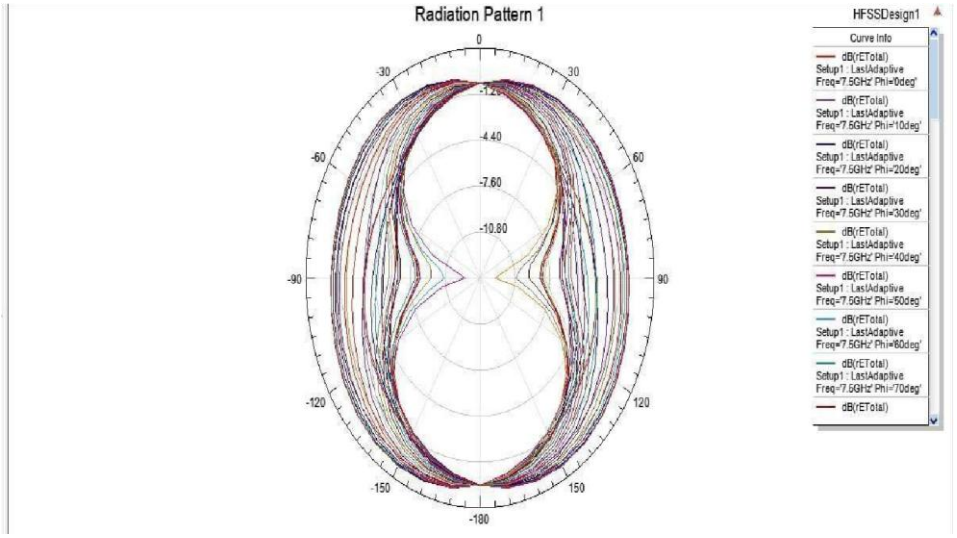


Fig 6: Radiation pattern

Envelope Correlation Coefficient (ECC) and Diversity Performance

ECC is one of the most important performance indicators of any MIMO antenna system. It measures how independently the antenna elements operate. A lower ECC indicates better diversity, improved system reliability, and higher communication quality.

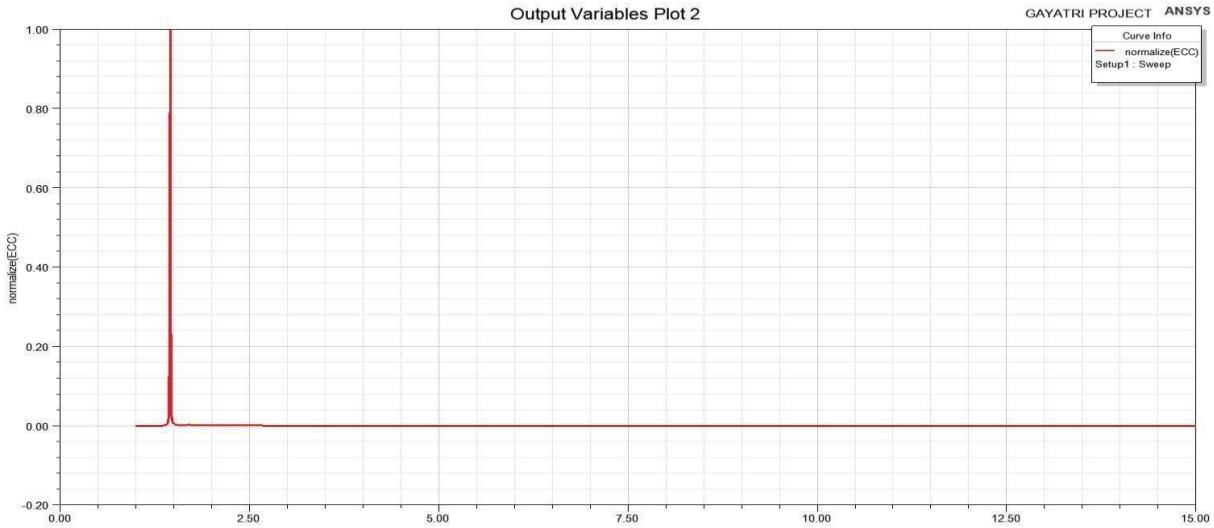


Fig 6: Envelope Correlation Coefficient (ECC)

Channel Capacity Loss

Channel capacity loss refers to the reduction in the maximum achievable data rate of a communication system compared to its ideal performance. In theory, channel capacity is defined by the Shannon–Hartley theorem, which relates capacity to bandwidth and signal-tonoise ratio (SNR). However, in practical systems, several non-ideal factors such as noise, interference, and hardware limitations prevent the system from reaching this theoretical limit. As a result, the actual data transmission rate is always lower than the ideal capacity.

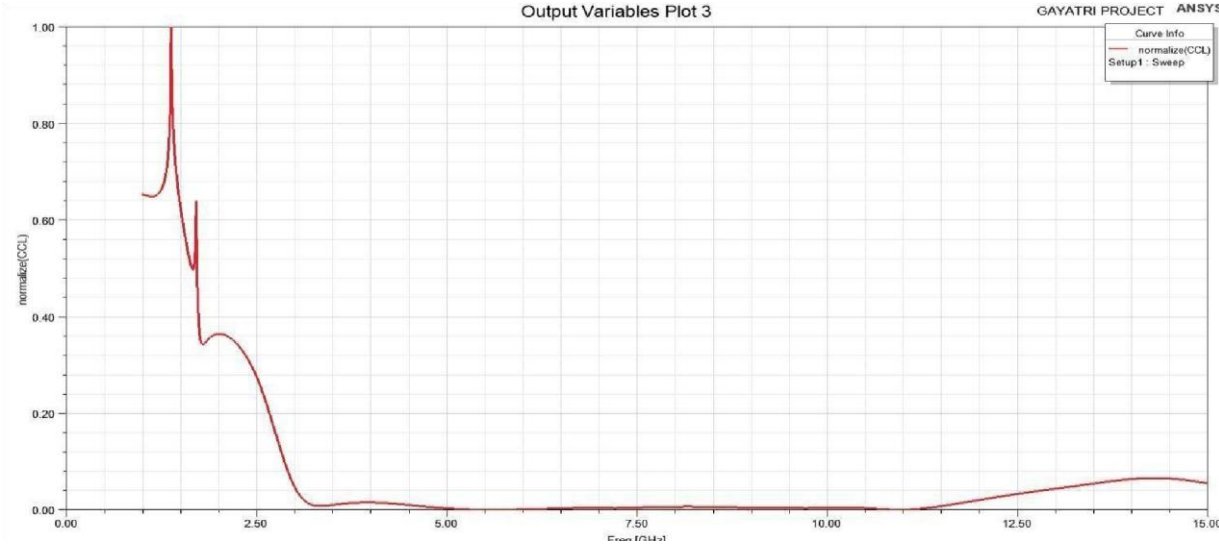


Fig 7:Channel Capacity Loss

Diversity Gain

Diversity Gain (DG) is a parameter in wireless communication that shows how much signal reliability improves when multiple antennas are used in a MIMO system. It represents the system's ability to reduce the effects of fading, noise, and signal fluctuations by combining signals from different antenna paths.

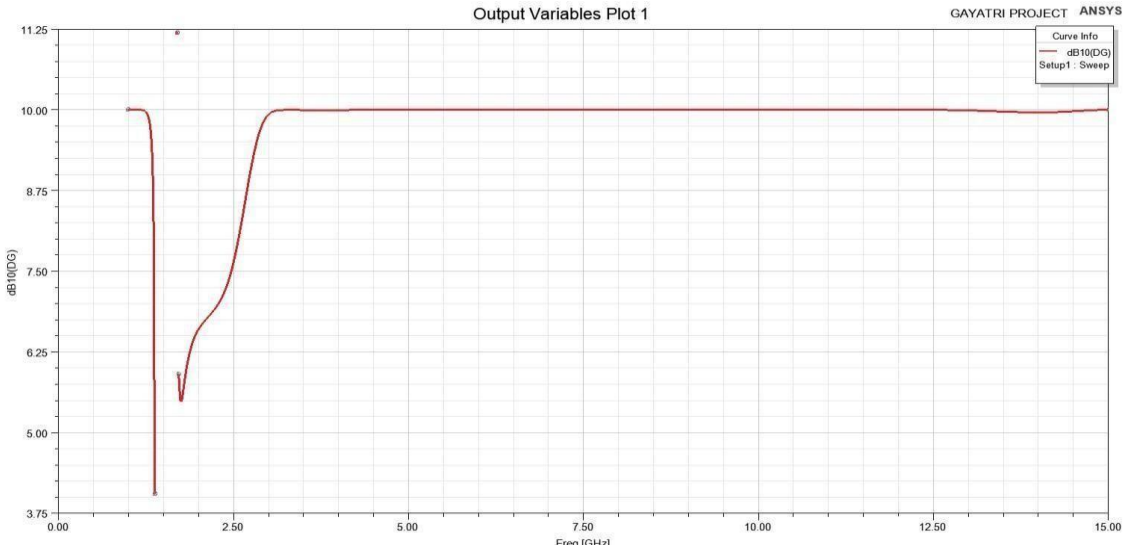


Fig 8: Diversity Gain Conclusion

The aim of this project was to design a compact Ultra-Wideband (UWB) MIMO antenna capable of supporting high-speed communication while maintaining excellent isolation, low mutual coupling, and strong diversity performance. The proposed design successfully meets all these objectives by combining two key innovations:

1. A staircase-shaped radiating structure, which enhances bandwidth through multiresonant behaviour. 2.

A comb-line ground structure, which effectively suppresses surface currents and reduces mutual coupling.

Simulation results confirmed that the antenna achieves wide impedance bandwidth from 3.1 GHz to 10.6 GHz, offering complete coverage of the FCC-defined UWB band. This verifies that the staircase radiator efficiently introduces multiple resonances that combine to produce stable ultra-wideband operation.

Additionally, the comb-line ground structure significantly enhances isolation, achieving S21 < –15 dB across the entire band, with even stronger isolation at mid-band frequencies. This demonstrates that the proposed decoupling technique is highly effective for compact MIMO designs.

The radiation characteristics remain stable throughout the operating band, with omnidirectional Hplane patterns and dipole-like E-plane patterns, ensuring reliable coverage in practical scenarios. The gain performance (2–5.5 dBi) and radiation efficiency (>70%) further confirm the antenna's suitability for portable and embedded UWB wireless devices. Most importantly, the antenna exhibits very low ECC values

(< 0.01), indicating excellent diversity gain and minimal channel correlation—critical for MIMO performance.

Overall, the proposed antenna demonstrates that high-speed UWB–MIMO performance can be achieved even on a low-cost FR4 substrate when the geometry is carefully optimized. The combination of compact size, high isolation, wide bandwidth, and strong diversity behaviour makes this antenna a suitable candidate for modern UWB applications including shortrange communication, real-time localization, biomedical sensing, and high-data-rate IoT.

Parameter	Achieved	Expected
Substrate Material	Rogers RT/duroid	Low-loss substrate required
Operating Bandwidth (S11)	3.1-10.6 GHz	Must cover UWB band
Return Loss (S11)	< -10 dB	< -10 dB
Isolation (S21)	< -15 to -20 dB	> -15 dB
ECC	< 0.01	< 0.1
Gain	2-6 dBi	Moderate expected
Efficiency	> 80%	> 70%
Radiation Pattern	Stable omni	Stable required
Diversity Gain	~10 dB	~10 dB ideal
Channel Capacity Loss	< 0.4 bits/s/Hz	< 0.4 bits/s/Hz

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